

A. To all Reviewers

Please allow us to restate our contributions as follows:

- We pioneeringly introduce focal tokens and token compression from vision domain into lidar-based detection, making it possible to process long-term temporal sequences. While the inherent lack of semantics in point clouds hinders the establishment of local correlations among points, rendering class-guided[4] and supervised methods[5] invalid. **Focal tokens**, highly information-dense token sets that efficiently represent instances, can be learned from a global perspective with our proposed adaptive method, laying the foundation of our pipeline.
- We propose a more insightful unified spatio-temporal fusion strategy. The non-scalability of data and the inherent architecture greatly limit existing temporal detectors in integrating global information. In contrast, our Grouped Hierarchical Fusion processes the entire temporal sequences from a global perspective. It can effectively extract spatio-temporal features and capture long-term dependencies in a progressive manner, offering a new view for processing variable-length sequences.

Our FASTER adheres to the principles of simplicity and effectiveness without bells and whistles. Thorough and well-designed experiments validate our claims and demonstrate FASTER’s superiority. We sincerely admire and appreciate your careful reading of this rebuttal letter. We believe that our statement will improve your perspective.

B. To Reviewer 61GY

We appreciate and will address the issues you raised.

PTT[3] focuses solely on box-level modeling, overlooking the importance of points and heavily relying on proposal recall, fundamentally distinguishing it from FASTER. Tabs 2 and 4 quantitatively demonstrate the superiority of FASTER against PTT. Fig. 1(a) is the point-based architecture shared by MPPNet[1] and MSF[2], while Fig. 1(b) clearly displays our another core innovation: the unified spatio-temporal fusion strategy, which progressively condenses long temporal sequences into a highly information-dense one.

Fig. 2 of appendix has visualized the evolution of token score distribution. In Fig. 6, we decode and visualize the focal tokens in the entire scene and are surprised to notice that historical tokens contribute more to moving instances.

The Intra-Group Fusion preserves temporal information, preventing our model from degenerating into a simple stacking. The dual-decoder introduces a learnable balance between the current and historical frames. The results in Tab 5 can sufficiently demonstrate their effectiveness, given the slight differences between SOTA detectors (see Tab 2).

We have provided the results on the Waymo test set in Tab 3 (verifiable on the official leaderboard, barring minor naming differences).

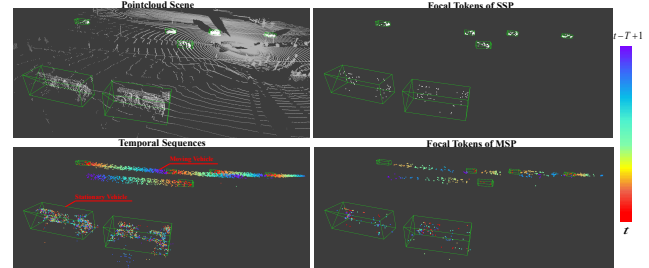


Figure 6. Visualization of Focal Tokens in a entire scene.

C. To Reviewer RbQ5

Thank you for your professional summary and recognition of our work. Now, we will answer your other questions.

In fact, motion features are inspired by MSF[2], significantly affecting performance (about 1.2% L2 mAPH). FASTER can flexibly handle sequences of any length with adaptive grouping. 16 frames offer the best efficiency-performance balance, though extending to 64 frames also improves accuracy (about 0.4%), demonstrating FASTER’s scalability. Both results have been added to the main text.

The sequence training you mentioned is very valuable. FASTER can indeed be trained in a streaming manner, but it would disrupt the data randomness. Unlike StreamPETR[6] storing historical queries and features, focal points only make up 1% of the entire ones (see Tab 1), requiring just 10GB of memory in total, which is fully supported. Moreover, our staged training and EPA effectively address this issue through simple focal points updates and storage.

D. To Reviewer eFb4

Thank you very much for your constructive feedback. In the following, we will address some of the concerns you raised.

In line 251, we explicitly state that the term **Adaptive** refers to that focal tokens are learned by the network rather than manually defined. Tab 3 of appendix has shown the impact of focal tokens, even an extremely small number of which (30%) can obtain competitive performance.

We have rephrased our contributions both in rationale and methodology, and clearly defined the core concept of “focal tokens”. We believe that mere “downsampling” can not accurately capture the essence of our work. Actually, we experimented with various token compression methods, but the adaptive manner we propose is the simplest and most effective, making it more universal and practical.

We sincerely appreciate your pointing out our weakness in expression. In fact, Ad-MHSA is introduced independently and out of context. As stated in lines 197 and 302, 4K points are sampled from each current proposal’s cylindrical region, and K points from historical proposals. We have improved the clarity of this section inspired by you.

References

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- [2] Chenhang He, Ruihuang Li, Yabin Zhang, Shuai Li, and Lei Zhang. Msf: Motion-guided sequential fusion for efficient 3d object detection from point cloud sequences. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5196–5205, 2023. [1](#)
- [3] Kuan-Chih Huang, Weijie Lyu, Ming-Hsuan Yang, and Yi-Hsuan Tsai. Ptt: Point-trajectory transformer for efficient temporal 3d object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14938–14947, 2024. [1](#)
- [4] Youwei Liang, Chongjian Ge, Zhan Tong, Yibing Song, Jue Wang, and Pengtao Xie. Not all patches are what you need: Expediting vision transformers via token reorganizations. *arXiv preprint arXiv:2202.07800*, 2022. [1](#)
- [5] Yongming Rao, Wenliang Zhao, Benlin Liu, Jiwen Lu, Jie Zhou, and Cho-Jui Hsieh. Dynamicvit: Efficient vision transformers with dynamic token sparsification. *Advances in neural information processing systems*, 34:13937–13949, 2021. [1](#)
- [6] Shihao Wang, Yingfei Liu, Tiancai Wang, Ying Li, and Xiangyu Zhang. Exploring object-centric temporal modeling for efficient multi-view 3d object detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 3621–3631, 2023. [1](#)